Combining gives

If , the following equation must have two distinct roots

in , for a suitable constant . The function must therefore be

nonmonotonic on ; and cannot both be increasing or

both decreasing. As grows, if rises, must fall; if falls,

must rise. Thus and have the same sign, giving

For , is zero at and . Hence on

has an extremum there. Thus has at least two distinct roots for a suitable .

Likewise for . Satisfying either of is necessary and sufficient.

Scoring (points): each ; ; ; and the

argument below them: points in total.

Method

In the figure, repel with ; attractions from are . Their

centripetal forces are ; distances to are . Similar triangles give

Dividing the two equations gives

The given conditions imply